

Sprint 3 Review Report

Project: Music Course Platform

Sprint: Sprint 3 (Week 6-7)

Date: March 1, 2026

Scrum Master: Su Wai Phyoe

Team Members: Luo Ying, Lu Liu, Su Wai Phyoe, Yicheng Chen

Sprint Goal

Extend functional prototype with teacher profile management, booking system, Jenkins CI/CD pipeline, Docker containerization, and enhanced UI for production readiness.

Key Objectives

- Implement Teacher Profile Feature and Booking System workflow
- Integrate Jenkins CI/CD Pipeline with automated testing and Docker containerization
- Enhance user interface with improved dashboards and comprehensive test coverage

Completed Tasks

- **Teacher Profile & Search** – Completed by Luo Ying & Lu Liu
Figma prototypes, teacher profile system with instrument search, JavaFX dashboard integration.
- **Booking System & Dashboard** – Completed by Liu Lu & Luo Ying
Complete booking workflow, time slot selection, learner dashboard, database schema updates.
- **Jenkins CI/CD Integration** – Completed by Yicheng Chen & Su Wai Phyoe
Jenkins installation, Jenkinsfile creation, automated pipeline with JaCoCo integration.
- **Docker & Testing** – Completed by Lu Liu, Su Wai Phyoe & Chen Yicheng
Multi-stage Dockerfile, local validation, enhanced unit testing suite with BookingDAOTest.
- **Time Slot Management** – Completed by Yicheng Chen & Luo Ying
Scheduling system, availability management, LoginController navigation updates.

Total Points Completed: 20/20 points (100%)

Demo Summary

- **Teacher System** – Complete profiles with search functionality
- **Booking Workflow** – End-to-end lesson scheduling
- **Jenkins CI/CD** – Automated pipeline (41% coverage, 48 unit tests)
- **Docker Image** – Containerized application ready for deployment
- **Enhanced UI** – Professional JavaFX dashboards

What Went Well

- All planned functionality delivered with effective DevOps integration
- Strong team collaboration through 6 productive meetings
- Successful Figma-to-JavaFX translation with robust testing framework

What Could Be Improved

- Frontend-backend integration coordination
- Environment consistency challenges
- Technical documentation enhancement needed

Time Spent During Sprint 3

Team Member	Hours	Role	Main Activities
Luo Ying	30h	Frontend Developer, UI/UX	Figma design, Database, teacher profile & search
Lu Liu	32h	Backend Developer, DevOps	Backend architecture, booking system, Database
Su Wai Phye	30h	Scrum Master, DevOps	Unit Testing, Jenkins CI/CD and Docker image
Yicheng Chen	32h	Backend Developer, DevOps	Time slot management, Dashboard, Docker

Technical Challenges and Solutions

- **JavaFX Compatibility** – Resolved macOS issues by upgrading to JavaFX 21.0.2
- **Database Case Sensitivity** – Fixed table naming inconsistencies
- **Jenkins Configuration** – Overcame working directory issues with proper Maven setup
- **Frontend Design Translation** – Successfully converted Figma designs to functional JavaFX interfaces

Conclusion

Sprint 3 achieved all objectives, delivering comprehensive teacher profile management, functional booking workflow, and robust CI/CD pipeline. Strong team collaboration and effective Trello coordination positioned the application for final sprint polish and production readiness.

Trello: <https://trello.com/b/B5AJ4wIm/sep1musiccourseplatform-sprint3>

GitHub: <https://github.com/chenyicheng1998/MusicCoursePlatform>